

# Model Fit

Money Ballers

2025-11-23

```
pacman::p_load(arrow, lme4, lmerTest, performance, broom, broom.mixed, patchwork,
               viridis, knitr, jtools, ggeffects, ggdist, gt, tidyverse)

options(scipen = 999)

# Set theme for plots
theme_set(theme_minimal())

# Setting repository path
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"
repo_path <- "D:/repos/Moneyball_FC/inputs/"

# Reading data
dat <- read_parquet(str_c(repo_path, "player_df.parquet"))
#dat <- read_parquet("player_df.parquet")
```

## Time Centering

- **Year:** Centering at 2013=0 makes the intercept interpretable as the expected value at the study's start and avoids numerical issues that can arise with large year values (2013-2024). Also, including  $\text{year}^2$  will allow us to study the (de)acceleration of the market over the decades.
  - Q: How does market value change annually compared to the 2013 baseline?

## Grand Mean Centering

- **Age:** Market value is determined by biological career stage (e.g., “is Messi 29 and in his prime?” vs “is Messi 19 and a prospect?”). It does *not* matter if you are the oldest guy

on a young team (relative age); the market still sees you as 29. We include the linear term (**age\_c**) to catch the general trend and the squared term (**age\_sq**) to capture the peak and decline curve.

- Q: How does market value change as a player moves through their biological career arc (from prospect to prime to veteran)?
- **Foreign Player Proportion:** Since no club has 0% foreign players (lowest is 28%), we shift the zero point to the population average (Grand Mean) of internationalization. The intercept therefore represents a club with ‘typical’ diversity for the league.
  - Q: Does having a *higher proportion of foreign players than the average club* increase market value?
- **Height:** Height is a static physical trait. The market values a 195cm defender because he is objectively tall, not because he is taller than his teammates. Grand mean centering allows us to interpret the coefficient as “the premium for being 1cm taller than the average soccer player. Keeps the intercept interpretable as the value of an “average height” player.
  - Q: Does being taller than the average player in the league increase market value?

### Year centering

- **UEFA Coefficient** = UEFA coefficients suffer from “grade inflation” or calculation changes over decades. A score of 15 in 2013 might mean something different than 15 in 2024. Centering by year isolates relative prestige, how dominant the club was that season compared to peers. It also removes the confounding effect of time-trends in the coefficient system itself.
  - Q: Does playing for a club that is currently outperforming the European average boost a player’s value?

### Year-Club mean centering (relative to teammates)

- **minutes\_played\_c:** Market value is driven by a player’s status within their team. This metric isolates whether a player is a “Starter” (positive value) or a “Bench Player” (negative value) relative to their specific squad in that specific season. It removes the noise of “teams that rotate more” vs “teams that don’t,” focusing purely on the player’s hierarchy in the squad.
  - Q: Does being a key player (playing more than the average teammate) increase market value?

### Position centering:

- **Goals, Assists:** A goal scored by a striker is expected; a goal scored by a defender is rare and highly valuable. If we leave **goals** and **assists** as raw counts, the model treats 1 goal the same for everyone, potentially undervaluing defensive contributions. We will group by position as performance expectations vary entirely by role. When we center within the position group, we transform the variable from a raw count into a “performance above expectation” metric. This interprets as the effect of scoring above position expectations.
  - Q: Does scoring more goals than the average player in their position increase market value?” (This distinguishes a goal-scoring defender from an underperforming striker).

### Non-centered fixed effects:

#### *Performance Metrics*

- **Red and Yellow card:** These variables have a meaningful 0 interpretation, and since our outcome is log transformed, these uncentered variables allow us to make percentage interpretations (i.e., one additional red card is associated with beta% change in market value).
  - Q: What is the direct market penalty for receiving one additional red/yellow card?
- **Position:** Reference group will be attackers.
  - Q: How much less is a defender valued compared to an attacker, holding performance constant?
- **Region:** Reference group European & Central Asia.
  - Q: What is the market premium for players in the ‘Europe’ region compared to South America?
- **Citizen:** Reference group is non-citizen of the country where the club is registered. The coefficient represents the simple difference in means between the two groups. A coefficient of +0.10 means “Citizens are valued 10% higher than Non-Citizens, all else held equal.”
  - Q: Is there a homegrown premium’ (or discount) for players who hold citizenship in the club’s country compared to foreign players?

#### Random Intercept and slope

- Players: Yearly market values are nested within players so this makes them a clustering variable.
  - Q: After controlling for goals and age, how much variance in value is simply due to who the player is (marketing value/reputation)?
- Club: Players are nested within groups, but they can also be members of multiple clubs throughout their career making this a secondary clustering variable.
  - Q: How much of the variation in player value is due to the club they play for ('Real Madrid Effect')?
- Year: Without this random slope, the model assumes a one-size-fits-all growth rate (parallel lines). By including it, we allow for the possibility that some clubs are incubators (where values skyrocket year-over-year) while others are stagnant (where values grow slowly or decline), even after controlling for player performance. The theoretical explanation is that elite clubs (e.g., Real Madrid, Manchester City) benefit from massive commercial exposure and "brand halos" that cause their players' values to inflate faster than the market average. Conversely, smaller clubs may see their players' values strictly tied to performance without the "inflationary" bonus of the club's brand over time. Without this random slope, we may underestimate the standard errors for time varying fixed effects.
  - Q: Does the rate of growth in player market value differ significantly between clubs?

### **Why cross-classified model?**

Players can change teams over their span of their careers meaning that their player context can change. They may be a star player in one team and a bench player on a different team that they switch to. In a cross-classified model, the player is the constant unit (i), but the club (j) changes over time (t). If player (I) plays for club (j1) in year (t0) but switches to club (j2) in year (t1) the fixed effect in the model, such as minutes played, updates the players status, as opposed to having static or average predictors. Club attributes, such as foreign players and UEFA scores, ensures that when player move clubs, the model swaps the values of  $X_{j1}$ , for  $X_{j2}$ . Player's characteristics, for instance age and height, move with the player across time.

### **Assumptions about the data:**

We log transform market value to 2024 USD because financial data follows a pareto distribution rather than a normal distribution. If we keep the outcome on the raw scale, our residuals will be extremely large for high value players, and small for low value players violating the assumption of homoscedasticity. Log transforming the outcome compresses the extreme outliers

by pulling the long right tail to try and make the distribution normal. We handle exchange rate fluctuation by using historical exchange rates to convert our values to USD. We also adjust market value to 2024 USD because 10 million in 2013 is not the same as in 2024 due to inflation, and by keeping the raw values we would lose real economic value. Therefore, this adjustment allows us to compare real economic power of a player in 2013 versus 2014 without the noise of currency fluctuation.

We assume that variables such as height and positions are time invariant for the duration of this study. At the club level, we assume mutually exclusiveness in that players can only belong to one club in a given year. Although a player could have switched clubs mid-season, we keep this constant for model simplicity. We assume that the relationship between age and log-market value is non-linear and parabolic so we introduce a quadratic term on age. Because we excluded observations with missing data (listwise deletion), we assume the data are Missing Completely at Random (MCAR) or that the proportion of missing cases (~2%) is negligible and does not introduce systematic bias.

Table 1: Data Structure for fixed effects, except year as it is a random slope

| Variable Type          | Variable                | Centering Strategy/<br>Form Factor | Justification                                                                                                                    |
|------------------------|-------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Time                   | Season Year             | Min-Centering (+ Quadratic)        | Establishes a meaningful starting intercept (2013 =0)                                                                            |
| Contextual, Club       | UEFA Coefficient        | Year-Mean Centering                | Isolates relative prestige; removes historical inflation                                                                         |
|                        | Foreign Players %       | Grand-Mean Centering               | Moves baseline to “Average Club” (since 0% is unlikely)                                                                          |
| Player Characteristics | Age                     | Grand-Mean (+ Quadratic)           | Captures absolute career stage (Prospect → Prime → Veteran)                                                                      |
|                        | Height                  | Grand-Mean                         | Captures premium for absolute physical size                                                                                      |
|                        | Position                | Reference group, attacker          | Attackers tend to be valued more on average                                                                                      |
| Player Demographic     | Citizen of Club Country | None (Raw binary 0/1)              | Keeps “Non-Citizen” as clear reference group. Separates individual status from club composition ( <code>foreign_players</code> ) |

| Variable Type      | Variable                            | Centering Strategy/<br>Form Factor | Justification                                                           |
|--------------------|-------------------------------------|------------------------------------|-------------------------------------------------------------------------|
| Player Performance | Region of origin                    | Reference group,<br>Europe         | Control variable for<br>where player is from                            |
|                    | Goals                               | Position-Centering                 | Preserves natural 0<br>and marginal value of<br>1 goal interpretation   |
|                    | Assists                             | Position-Centering                 | Preserves natural 0<br>and marginal value of<br>1 assist interpretation |
|                    | Red Card                            | raw                                | Meaningful 0<br>interpretation                                          |
|                    | Yellow Card                         | raw                                | Meaningful 0<br>interpretation                                          |
|                    | Average minutes<br>played in season | Year-Mean<br>Centering, z-score    | Isolates whether<br>player is a starter or<br>bench player              |

```

player_df <- dat |>
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    # reference groups
    position = fct_relevel(position, "Attack"),
    Region = fct_relevel(Region, "Europe & Central Asia"),
    # outcome on log scale
    usd_revenue_2024_log = log(usd_revenue_2024),
    # season year
    # convert season_year to 0, 1, 2, etc.
    season_year_c = season_year - min(season_year),
    # grand mean centering
    foreign_players = foreign_players - mean(foreign_players, na.rm=TRUE),
    age_c = Age - mean(Age, na.rm=TRUE),
    height_c = height_in_cm - mean(height_in_cm, na.rm=TRUE)
  ) |>
  # year mean centering

```



Figure 1: Cross-Classified Data Structure: Players move between clubs over time.

```

group_by(season_year_c) |>
mutate(uefa_coefficient = uefa_coefficient - mean(uefa_coefficient, na.rm=TRUE),
       minutes_played = (minutes_played - mean(minutes_played, na.rm=TRUE)) /
       sd(minutes_played, na.rm = TRUE)) |>
# position centering
group_by(position) |>
mutate(goals = goals - mean(goals, na.rm=TRUE),
       assists = assists - mean(assists, na.rm=TRUE)) |>
ungroup()

```

- After checking missing values, we saw no more than 108 missing cases for Region, 103 for citizen of club country, and we decide to drop these cases as there are few and we are left with 7,023 cases.

```

# Which variable has missing value
player_df |>
  reframe(across(everything(), ~sum(is.na(.)))) |>
  select_if(~ . > 0) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing Cases") |>
  slice(1:5) |>
  arrange(desc(`Missing Cases`)) |>
  kable()

```

| Variable               | Missing Cases |
|------------------------|---------------|
| country_of_birth       | 108           |
| country_of_citizenship | 103           |
| height_in_cm           | 36            |
| date_of_birth          | 2             |
| Age                    | 2             |

```

clean_player_df <- player_df |> drop_na()

```

- After examining the distribution of log market values across clubs, it is evident that the boxplots differ in both central tendency and variability. This indicates that club is a meaningful factor in explaining variation in market values.

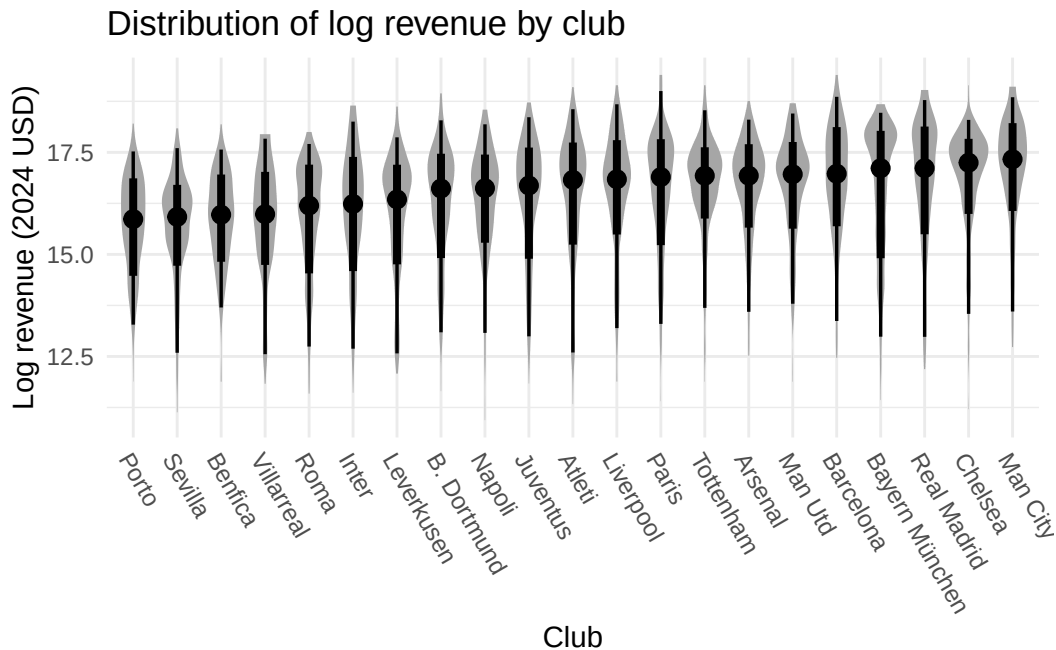
```

clean_player_df |>
  mutate(Club = fct_reorder(factor(Club), usd_revenue_2024_log, median)) |>
  ggplot(aes(x=Club, y=usd_revenue_2024_log, group=Club)) +
  stat_slabinterval(side = "both") +

```



```
theme(axis.text.x=element_text(angle = -60, hjust = 0)) +
labs(title = "Distribution of log revenue by club",
     y="Log revenue (2024 USD)")
```



## Linearity

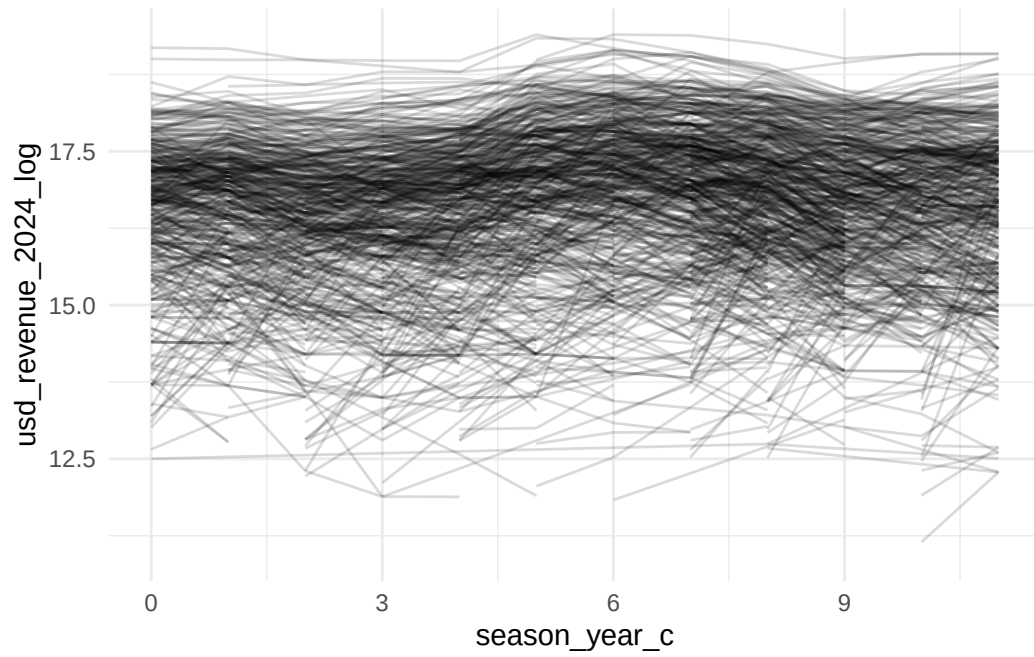
In the overall spaghetti plot group by `play_id`, the lines vary widely across players, showing considerable between-player heterogeneity in log market value. Within most players, the slopes appear mostly flat, but some of them change up and down dramatically.

In the overall spaghetti plot group by `club_id`, they are consistently show differences. This means between-club differences in market values are large, supporting the idea that club-level random intercepts are necessary in your multilevel model.

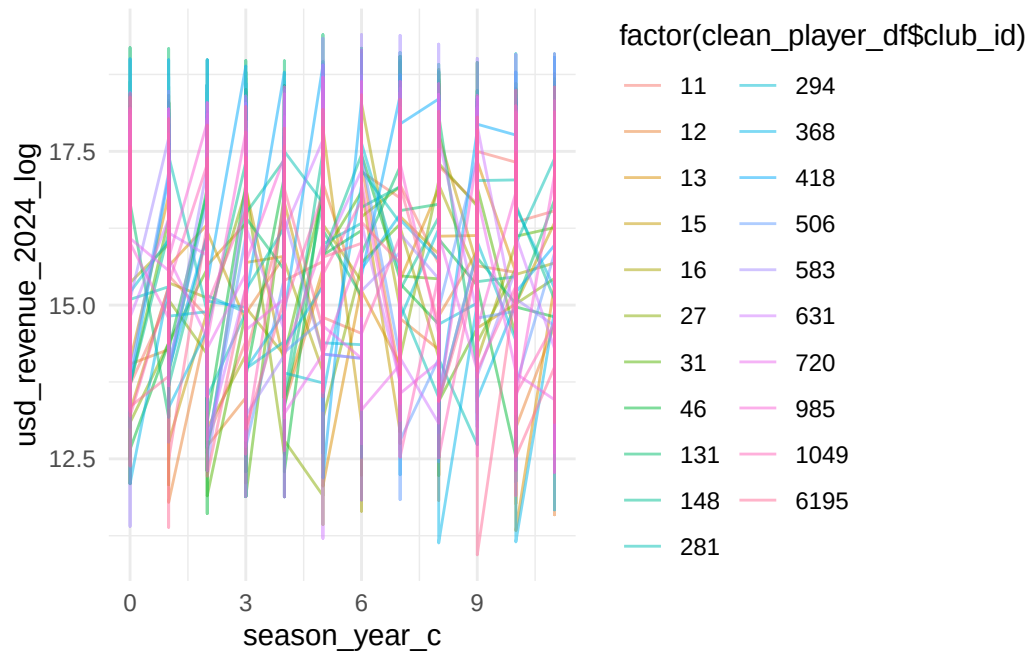
Besides, the clubs' average values change over time, indicating slopes might vary across clubs.

The loess smoother plot shows a non-linear averaged trend (curvature) in log-transformed market value (`usd_revenue_2024_log`) over centered season year (`season_year_c`). We can clearly see a peak at about season year 6 to 7, and the shape of the loess line is a wave pattern. Thus, we can try to add polynomial terms in our model to better explain the data.

```
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
               group = player_id), color = "black", alpha = 0.15)
```

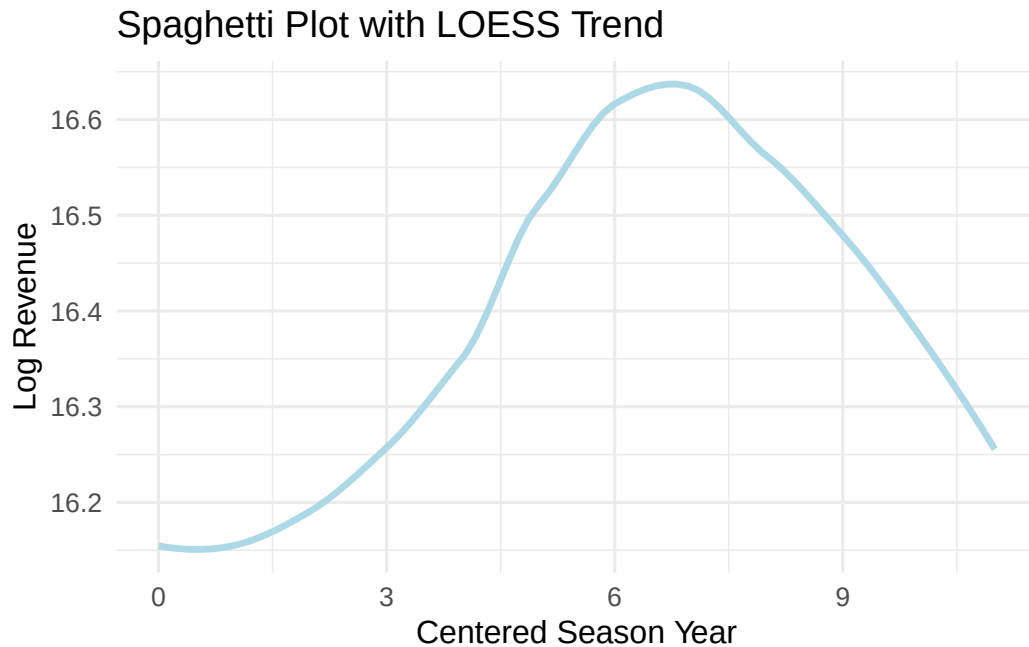


```
ggplot(clean_player_df) +  
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,  
                group = club_id, color = factor(clean_player_df$club_id)),  
            alpha = 0.5)
```



```
ggplot(clean_player_df, aes(x = season_year_c,
                           y = usd_revenue_2024_log)) +

  # LOESS smoother (overall trend)
  geom_smooth(method = "loess",
              color = "lightblue", size = 1.2, se = FALSE) +
  labs(x = "Centered Season Year",
       y = "Log Revenue",
       title = "Spaghetti Plot with LOESS Trend") +
  theme_minimal(base_size = 12)
```



## Modeling Process Plan

- Model 1: Random Intercepts only to examine ICC
- Testing whether a random slope on year is needed.
  - Model 2.1: Random intercepts + fixed effect of centered season\_\_year
  - Model 2.2: Adding quadratic terms for season year
  - Model 2.3: Adding cubic term for season year
  - Model 2.4 Adding random slope of season\_\_year\_\_c
  - Model 2.5: Adding quadratic term of season\_\_year as a random slope
- Adding level fixed effects
  - Establish whether age quadratic term is needed
    - \* Model 3.1: Adding player covariates {Height + Position + Citizen Status + Region + Age + Goals + Assists + Red Cards + Yellow cards + Minutes played}

- \* Model 3.2: Adding the quadratic term for age
- Model 4: Adding club covariates {UEFA coefficients + Foreign Players}
- Compare model fit
- Model diagnostic
- Choose final model

### Model 1, intercepts only

```
#Using bobyqa to handle the complex random effects structure and avoid gradient warnings
strict_control <- lmerControl(optimizer = "bobyqa",
                              optCtrl = list(maxfun = 200000))

Model_1 <- lmer(
  usd_revenue_2024_log ~
    # random effects
    (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_1)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|                                       |            |
|---------------------------------------|------------|
| AIC                                   | 20105.1893 |
| BIC                                   | 20132.6171 |
| Pseudo-R <sup>2</sup> (fixed effects) | 0.0000     |
| Pseudo-R <sup>2</sup> (total)         | 0.7799     |

- We can see that player\_id accounts for about 75% of market value while club\_id accounts for 3%.

| Fixed Effects |         |        |          |         |        |
|---------------|---------|--------|----------|---------|--------|
|               | Est.    | S.E.   | t val.   | d.f.    | p      |
| (Intercept)   | 15.8205 | 0.0691 | 228.8952 | 26.9891 | 0.0000 |

p values calculated using Satterthwaite d.f.

| Random Effects |             |           |
|----------------|-------------|-----------|
| Group          | Parameter   | Std. Dev. |
| player_id      | (Intercept) | 1.3209    |
| club_id        | (Intercept) | 0.2837    |
| Residual       |             | 0.7176    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7455 |
| club_id            | 21       | 0.0344 |

## Model 2.1: Adding fixed effect of year

```
Model_2.1 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.1)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|                                       |            |
|---------------------------------------|------------|
| AIC                                   | 20052.8025 |
| BIC                                   | 20087.0872 |
| Pseudo-R <sup>2</sup> (fixed effects) | 0.0048     |
| Pseudo-R <sup>2</sup> (total)         | 0.7907     |

| Fixed Effects |         |        |          |           |        |
|---------------|---------|--------|----------|-----------|--------|
|               | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)   | 15.9908 | 0.0723 | 221.3121 | 32.9064   | 0.0000 |
| season_year_c | -0.0310 | 0.0038 | -8.1131  | 6524.0329 | 0.0000 |

p values calculated using Satterthwaite d.f.

| Random Effects |             |           |
|----------------|-------------|-----------|
| Group          | Parameter   | Std. Dev. |
| player_id      | (Intercept) | 1.3441    |
| club_id        | (Intercept) | 0.2822    |
| Residual       |             | 0.7088    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7563 |
| club_id            | 21       | 0.0333 |

## Model 2.2 Adding quadratic term

- Notice that the linear relationship of year is no longer significant.

```
Model_2.2 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 2, raw = TRUE) +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.2)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|                                       |            |
|---------------------------------------|------------|
| AIC                                   | 20046.9776 |
| BIC                                   | 20088.1192 |
| Pseudo-R <sup>2</sup> (fixed effects) | 0.0058     |
| Pseudo-R <sup>2</sup> (total)         | 0.7913     |

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 15.9164 | 0.0741 | 214.8843 | 36.5518   | 0.0000 |
| poly(season_year_c, 2, raw = TRUE)1 | 0.0187  | 0.0118 | 1.5836   | 5756.8595 | 0.1133 |
| poly(season_year_c, 2, raw = TRUE)2 | -0.0045 | 0.0010 | -4.4510  | 5650.5050 | 0.0000 |

p values calculated using Satterthwaite d.f.

| Random Effects |             |           |
|----------------|-------------|-----------|
| Group          | Parameter   | Std. Dev. |
| player_id      | (Intercept) | 1.3434    |
| club_id        | (Intercept) | 0.2818    |
| Residual       |             | 0.7076    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7568 |
| club_id            | 21       | 0.0333 |

**Compare Model 2.1 and 2.2 to determine if the quadratic term is needed.**

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the quadratic term is needed for this data.

```
anova(Model_2.1, Model_2.2, test = "LRT") |> gt()
```



| npars | AIC      | BIC      | logLik    | -2*log(L) | Chisq    | Df | Pr(>Chisq)     |
|-------|----------|----------|-----------|-----------|----------|----|----------------|
| 5     | 20039.98 | 20074.27 | -10014.99 | 20029.98  | NA       | NA | NA             |
| 6     | 20022.19 | 20063.33 | -10005.10 | 20010.19  | 19.78947 | 1  | 0.000008645778 |

### Model 2.3 Adding cubic term

- Now linear year trends is significant

```
Model_2.3 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.3)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|                                       |            |
|---------------------------------------|------------|
| AIC                                   | 20037.4036 |
| BIC                                   | 20085.4022 |
| Pseudo-R <sup>2</sup> (fixed effects) | 0.0067     |
| Pseudo-R <sup>2</sup> (total)         | 0.7930     |

### Compare Model 2.2 and 2.3 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the cubic term is needed for this data. After adding cubic terms,

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 15.9932 | 0.0756 | 211.5628 | 39.5941   | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0926 | 0.0248 | -3.7333  | 5460.0957 | 0.0002 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0220  | 0.0053 | 4.1535   | 5343.5767 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0016 | 0.0003 | -5.0949  | 5343.9721 | 0.0000 |

p values calculated using Satterthwaite d.f.

| Random Effects |             |           |
|----------------|-------------|-----------|
| Group          | Parameter   | Std. Dev. |
| player_id      | (Intercept) | 1.3458    |
| club_id        | (Intercept) | 0.2819    |
| Residual       |             | 0.7055    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7583 |
| club_id            | 21       | 0.0333 |

it yields a better fit. It cannot only explain the peak of the overall trajectory, but can also explain the up and down wave pattern over time of the trajectory.

```
anova(Model_2.2, Model_2.3, test = "LRT") |> gt()
```

## Model 2.4, Random slope

- To estimate a random slope ( $1 + \text{year\_centered} \mid \text{player\_id}$ ), the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the “player slope” from the “residual error,” causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

| npar | AIC      | BIC      | logLik     | -2*log(L) | Chisq    | Df | Pr(>Chisq)      |
|------|----------|----------|------------|-----------|----------|----|-----------------|
| 6    | 20022.19 | 20063.33 | -10005.096 | 20010.19  | NA       | NA | NA              |
| 7    | 19998.32 | 20046.32 | -9992.159  | 19984.32  | 25.87479 | 1  | 0.0000003642961 |

```
Model_2.4 <- lmer(
  usd_revenue_2024_log ~
    ## first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.4)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|     |            |
|-----|------------|
| AIC | 19974.7175 |
| BIC | 20036.4300 |

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 15.9823 | 0.0887 | 180.2113 | 28.0241   | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0920 | 0.0260 | -3.5333  | 1018.9716 | 0.0004 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0226  | 0.0052 | 4.3039   | 5311.3234 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0016 | 0.0003 | -5.2901  | 5311.7328 | 0.0000 |

p values calculated using Satterthwaite d.f.

### Model 2.4, Random slope + quadratic term

- We have a convergence problem, it seems that adding the quadratic term makes the model unidentifiable. We move forward with Models 2.2 and 2.3.

| Random Effects |               |           |
|----------------|---------------|-----------|
| Group          | Parameter     | Std. Dev. |
| player_id      | (Intercept)   | 1.3508    |
| club_id        | (Intercept)   | 0.3532    |
| club_id        | season_year_c | 0.0394    |
| Residual       |               | 0.6975    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7491 |
| club_id            | 21       | 0.0512 |

```
Model_2.5 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
    (1 | player_id) + (season_year_c + season_year_c^2 | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.5)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|     |            |
|-----|------------|
| AIC | 19974.7175 |
| BIC | 20036.4300 |

**Determine if the random slope on season year is needed**

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 15.9823 | 0.0887 | 180.2113 | 28.0241   | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0920 | 0.0260 | -3.5333  | 1018.9716 | 0.0004 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0226  | 0.0052 | 4.3039   | 5311.3234 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0016 | 0.0003 | -5.2901  | 5311.7328 | 0.0000 |

p values calculated using Satterthwaite d.f.

| Random Effects |               |           |
|----------------|---------------|-----------|
| Group          | Parameter     | Std. Dev. |
| player_id      | (Intercept)   | 1.3508    |
| club_id        | (Intercept)   | 0.3532    |
| club_id        | season_year_c | 0.0394    |
| Residual       |               | 0.6975    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7491 |
| club_id            | 21       | 0.0512 |

```
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced <- as.numeric(logLik(reduced_model))
  ll_full    <- as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  # Formula: -2 * (LL_reduced - LL_full)
  lrt_stat <- -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  # This tells us how many params we added
  df_reduced <- attr(logLik(reduced_model), "df")
  df_full    <- attr(logLik(full_model), "df")
  df_diff    <- df_full - df_reduced

  # 4. Calculate P-Value based on the difference
```

```

# If we added a slope + covariance, df_diff should be 2.
# The mixture is 0.5 * ChiSq(1) + 0.5 * ChiSq(2)

if (df_diff == 2) {
  p_val <- 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

  msg <- "Mixture of ChiSq(1) and ChiSq(2)"
} else if (df_diff == 1) {
  # If we added a parameter with NO covariance (e.g. diag structure)
  p_val <- 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

  msg <- "Mixture of ChiSq(0) and ChiSq(1)"
} else {
  # Fallback for non-standard comparisons
  p_val <- pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
  msg <- paste0("Standard ChiSq(", df_diff, ")")
}

# Output
cat("LRT Statistic:", round(lrt_stat, 4), "\n")
cat("Parameter Diff:", df_diff, "\n")
cat("Distribution: ", msg, "\n")
cat("P-Value:      ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(Model_2.3, Model_2.4)

```

```

LRT Statistic: 66.6861
Parameter Diff: 2
Distribution:   Mixture of ChiSq(1) and ChiSq(2)
P-Value:       < 0.0001

```

- We use a helper function to print the p-value for the LRT difference, and find a p-value  $< .0001$  and we reject the Null, (with  $.5 \times \chi^2(1) + .5 \times \chi^2(2)$  mixture) and conclude that there is strong evidence for club variation in market value trajectories and therefore we should keep this random slope in the model.

### Model 3.1: Adding fixed effects at level 1 with age as a linear relationship

- Age centered is not even significant. Same for red\yellow cards.

```
Model_3.1 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_3.1)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|     |            |
|-----|------------|
| AIC | 17469.2423 |
| BIC | 17640.6660 |

### Model 3.2. Adding age quadratic term

```
Model_3.2 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
```

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 16.5121 | 0.1020 | 161.8997 | 44.4734   | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0538 | 0.0226 | -2.3814  | 1148.4380 | 0.0174 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0222  | 0.0046 | 4.8478   | 5237.4683 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0018 | 0.0003 | -6.6889  | 5239.0873 | 0.0000 |
| height_c                            | 0.0140  | 0.0037 | 3.7216   | 1669.4917 | 0.0002 |
| positionDefender                    | -0.4310 | 0.0596 | -7.2306  | 1650.1235 | 0.0000 |
| positionGoalkeeper                  | -1.2404 | 0.0976 | -12.7037 | 1666.8564 | 0.0000 |
| positionMidfield                    | -0.2276 | 0.0595 | -3.8227  | 1677.8494 | 0.0001 |
| citizen_of_club_countryYes          | -0.3992 | 0.0376 | -10.6299 | 5729.2044 | 0.0000 |
| RegionEast Asia & Pacific           | -0.1817 | 0.2112 | -0.8605  | 1734.4279 | 0.3896 |
| RegionLatin America & Caribbean     | 0.2481  | 0.0634 | 3.9134   | 1796.5122 | 0.0001 |
| RegionMiddle East & North Africa    | 0.0833  | 0.2452 | 0.3396   | 1697.4147 | 0.7342 |
| RegionNorth America                 | -0.0053 | 0.2249 | -0.0234  | 1778.5037 | 0.9814 |
| RegionSub-Saharan Africa            | 0.0183  | 0.1038 | 0.1759   | 1732.4873 | 0.8604 |
| age_c                               | -0.0010 | 0.0040 | -0.2525  | 2354.7908 | 0.8007 |
| goals                               | 0.0139  | 0.0026 | 5.3168   | 5538.6861 | 0.0000 |
| assists                             | 0.0126  | 0.0036 | 3.4445   | 5075.1564 | 0.0006 |
| red_cards                           | -0.0187 | 0.0296 | -0.6338  | 4696.8271 | 0.5262 |
| yellow_cards                        | 0.0058  | 0.0041 | 1.4203   | 5406.1624 | 0.1556 |
| minutes_played                      | 0.5115  | 0.0174 | 29.4113  | 5605.9443 | 0.0000 |

p values calculated using Satterthwaite d.f.

| Random Effects |               |           |
|----------------|---------------|-----------|
| Group          | Parameter     | Std. Dev. |
| player_id      | (Intercept)   | 0.9464    |
| club_id        | (Intercept)   | 0.3686    |
| club_id        | season_year_c | 0.0329    |
| Residual       |               | 0.6145    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.6356 |
| club_id            | 21       | 0.0964 |



```
# random effects
(1 | player_id) + (season_year_c | club_id),
data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_3.2)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|     |            |
|-----|------------|
| AIC | 14507.0975 |
| BIC | 14685.3781 |

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 16.8498 | 0.0783 | 215.3074 | 82.3319   | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0557 | 0.0169 | -3.2907  | 2124.8495 | 0.0010 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0276  | 0.0035 | 7.9468   | 5036.1261 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0022 | 0.0002 | -10.8944 | 5037.4463 | 0.0000 |
| height_c                            | 0.0098  | 0.0036 | 2.6949   | 1799.8481 | 0.0071 |
| positionDefender                    | -0.4037 | 0.0578 | -6.9786  | 1786.1940 | 0.0000 |
| positionGoalkeeper                  | -1.0571 | 0.0947 | -11.1594 | 1797.7097 | 0.0000 |
| positionMidfield                    | -0.1905 | 0.0577 | -3.3014  | 1802.0409 | 0.0010 |
| citizen_of_club_countryYes          | -0.1898 | 0.0314 | -6.0394  | 6875.7653 | 0.0000 |
| RegionEast Asia & Pacific           | -0.4305 | 0.2041 | -2.1098  | 1836.2341 | 0.0350 |
| RegionLatin America & Caribbean     | 0.1908  | 0.0609 | 3.1332   | 1954.2127 | 0.0018 |
| RegionMiddle East & North Africa    | -0.0917 | 0.2374 | -0.3864  | 1814.6163 | 0.6992 |
| RegionNorth America                 | -0.0634 | 0.2169 | -0.2923  | 1856.8054 | 0.7701 |
| RegionSub-Saharan Africa            | 0.0659  | 0.1003 | 0.6570   | 1848.9110 | 0.5113 |
| poly(age_c, 2, raw = TRUE)1         | 0.0186  | 0.0037 | 4.9884   | 2295.0021 | 0.0000 |
| poly(age_c, 2, raw = TRUE)2         | -0.0242 | 0.0004 | -62.7957 | 6152.2162 | 0.0000 |
| goals                               | 0.0047  | 0.0020 | 2.3634   | 5169.1922 | 0.0181 |
| assists                             | 0.0095  | 0.0028 | 3.4497   | 4882.6031 | 0.0006 |
| red_cards                           | 0.0095  | 0.0222 | 0.4279   | 4678.4516 | 0.6687 |
| yellow_cards                        | -0.0018 | 0.0031 | -0.5826  | 5089.4375 | 0.5602 |
| minutes_played                      | 0.3528  | 0.0134 | 26.2596  | 5243.8360 | 0.0000 |

p values calculated using Satterthwaite d.f.

| Random Effects |               |           |
|----------------|---------------|-----------|
| Group          | Parameter     | Std. Dev. |
| player_id      | (Intercept)   | 0.9599    |
| club_id        | (Intercept)   | 0.2373    |
| club_id        | season_year_c | 0.0186    |
| Residual       |               | 0.4589    |

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7754 |
| club_id            | 21       | 0.0474 |

- It seems that we need the quadratic term for age in the model.

```
anova(Model_3.1, Model_3.2) |> gt()
```

#### Model 4: Adding club covariates

- The model with club covariates under performs compared to model 3.2, yet foreign players is significant. We generate a new model 4 dropping UEFA coefficients and test against 3.2 again.

```
Model_4 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
```

| npar | AIC      | BIC      | logLik    | -2*log(L) | Chisq    | Df | Pr(>Chisq) |
|------|----------|----------|-----------|-----------|----------|----|------------|
| 25   | 17345.33 | 17516.75 | -8647.664 | 17295.33  | NA       | NA | NA         |
| 26   | 14362.95 | 14541.23 | -7155.476 | 14310.95  | 2984.376 | 1  | 0          |

```

uefa_coefficient + foreign_players +
# random effects
(1 | player_id) + (season_year_c | club_id),
data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_4)

```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|     |            |
|-----|------------|
| AIC | 14520.4840 |
| BIC | 14712.4785 |

```
anova(Model_3.2, Model_4) |> gt()
```

- This time, model 4 is a better fit with only 1 club covariate. AIC is lower for Model 4, but BIC is higher for Model 3.2. We decide to keep Model 4 as it contains club contextual covariates information.

```

Model_4 <- lmer(
  usd_revenue_2024_log ~
  # first level fixed effect
  # Use poly() with raw=TRUE
  poly(season_year_c, 3, raw = TRUE) +
  # player fixed effects
  height_c + position + citizen_of_club_country + Region +
  poly(age_c, 2, raw = TRUE) +
  goals + assists + red_cards + yellow_cards + minutes_played +
  # club fixed effects
  foreign_players +
  # random effects
  (1 | player_id) + (season_year_c | club_id),
data = clean_player_df, REML = TRUE, control = strict_control)

```

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 16.8574 | 0.0793 | 212.4550 | 77.4322   | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0610 | 0.0172 | -3.5499  | 1979.2430 | 0.0004 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0283  | 0.0035 | 8.0967   | 5037.1907 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0023 | 0.0002 | -10.9692 | 5035.6950 | 0.0000 |
| height_c                            | 0.0098  | 0.0036 | 2.7041   | 1797.2098 | 0.0069 |
| positionDefender                    | -0.4040 | 0.0578 | -6.9914  | 1783.3530 | 0.0000 |
| positionGoalkeeper                  | -1.0566 | 0.0946 | -11.1657 | 1795.0980 | 0.0000 |
| positionMidfield                    | -0.1903 | 0.0576 | -3.3020  | 1799.2532 | 0.0010 |
| citizen_of_club_countryYes          | -0.1860 | 0.0315 | -5.9054  | 6884.1551 | 0.0000 |
| RegionEast Asia & Pacific           | -0.4307 | 0.2039 | -2.1129  | 1833.7205 | 0.0347 |
| RegionLatin America & Caribbean     | 0.1889  | 0.0608 | 3.1056   | 1951.9683 | 0.0019 |
| RegionMiddle East & North Africa    | -0.0936 | 0.2372 | -0.3948  | 1811.8833 | 0.6930 |
| RegionNorth America                 | -0.0659 | 0.2167 | -0.3040  | 1854.3493 | 0.7612 |
| RegionSub-Saharan Africa            | 0.0638  | 0.1002 | 0.6369   | 1846.5714 | 0.5243 |
| poly(age_c, 2, raw = TRUE)1         | 0.0184  | 0.0037 | 4.9545   | 2294.6335 | 0.0000 |
| poly(age_c, 2, raw = TRUE)2         | -0.0242 | 0.0004 | -62.7901 | 6163.0575 | 0.0000 |
| goals                               | 0.0048  | 0.0020 | 2.3857   | 5164.5854 | 0.0171 |
| assists                             | 0.0096  | 0.0028 | 3.4731   | 4878.2160 | 0.0005 |
| red_cards                           | 0.0087  | 0.0223 | 0.3907   | 4673.6600 | 0.6960 |
| yellow_cards                        | -0.0018 | 0.0031 | -0.5724  | 5084.8163 | 0.5671 |
| minutes_played                      | 0.3529  | 0.0134 | 26.2699  | 5239.3341 | 0.0000 |
| uefa_coefficient                    | -0.0011 | 0.0012 | -0.9810  | 4466.8449 | 0.3267 |
| foreign_players                     | 0.1995  | 0.0971 | 2.0533   | 1972.8895 | 0.0402 |

p values calculated using Satterthwaite d.f.

| Random Effects |               |           |
|----------------|---------------|-----------|
| Group          | Parameter     | Std. Dev. |
| player_id      | (Intercept)   | 0.9588    |
| club_id        | (Intercept)   | 0.2445    |
| club_id        | season_year_c | 0.0194    |
| Residual       |               | 0.4589    |

```
summ(digits = 4, Model_4)
```

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7728 |
| club_id            | 21       | 0.0502 |

| npair | AIC      | BIC      | logLik    | -2*log(L) | Chisq    | Df | Pr(>Chisq) |
|-------|----------|----------|-----------|-----------|----------|----|------------|
| 26    | 14362.95 | 14541.23 | -7155.476 | 14310.95  | NA       | NA | NA         |
| 28    | 14361.89 | 14553.88 | -7152.944 | 14305.89  | 5.064713 | 2  | 0.07947152 |

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|     |            |
|-----|------------|
| AIC | 14507.7590 |
| BIC | 14692.8966 |

```
anova(Model_3.2, Model_4) |> gt()
```

### Bonus round, Model 5 testing 2 levels only for player data.

- We drop the random intercept for club and move the random slope of season year to player and find that this is a worse fit than Model 4.
- Model 4 is our final model

```
Model_5 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    foreign_players +
```

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 16.8572 | 0.0790 | 213.4409 | 79.1128   | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0611 | 0.0172 | -3.5602  | 2024.2364 | 0.0004 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0283  | 0.0035 | 8.1128   | 5037.2537 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0023 | 0.0002 | -10.9890 | 5035.1017 | 0.0000 |
| height_c                            | 0.0098  | 0.0036 | 2.6981   | 1796.9647 | 0.0070 |
| positionDefender                    | -0.4040 | 0.0578 | -6.9923  | 1783.2364 | 0.0000 |
| positionGoalkeeper                  | -1.0573 | 0.0946 | -11.1737 | 1794.7986 | 0.0000 |
| positionMidfield                    | -0.1905 | 0.0576 | -3.3047  | 1799.1086 | 0.0010 |
| citizen_of_club_countryYes          | -0.1861 | 0.0315 | -5.9085  | 6884.2560 | 0.0000 |
| RegionEast Asia & Pacific           | -0.4316 | 0.2038 | -2.1172  | 1833.5081 | 0.0344 |
| RegionLatin America & Caribbean     | 0.1889  | 0.0608 | 3.1055   | 1951.8629 | 0.0019 |
| RegionMiddle East & North Africa    | -0.0939 | 0.2372 | -0.3958  | 1811.7669 | 0.6923 |
| RegionNorth America                 | -0.0666 | 0.2166 | -0.3073  | 1854.1815 | 0.7586 |
| RegionSub-Saharan Africa            | 0.0635  | 0.1002 | 0.6339   | 1846.4296 | 0.5262 |
| poly(age_c, 2, raw = TRUE)1         | 0.0184  | 0.0037 | 4.9562   | 2294.5304 | 0.0000 |
| poly(age_c, 2, raw = TRUE)2         | -0.0242 | 0.0004 | -62.7837 | 6159.9545 | 0.0000 |
| goals                               | 0.0047  | 0.0020 | 2.3749   | 5166.6298 | 0.0176 |
| assists                             | 0.0096  | 0.0028 | 3.4706   | 4879.7043 | 0.0005 |
| red_cards                           | 0.0088  | 0.0223 | 0.3960   | 4674.9908 | 0.6921 |
| yellow_cards                        | -0.0018 | 0.0031 | -0.5770  | 5086.4328 | 0.5640 |
| minutes_played                      | 0.3530  | 0.0134 | 26.2763  | 5241.1391 | 0.0000 |
| foreign_players                     | 0.1994  | 0.0971 | 2.0535   | 1948.7098 | 0.0402 |

p values calculated using Satterthwaite d.f.

| Random Effects |               |           |
|----------------|---------------|-----------|
| Group          | Parameter     | Std. Dev. |
| player_id      | (Intercept)   | 0.9587    |
| club_id        | (Intercept)   | 0.2420    |
| club_id        | season_year_c | 0.0192    |
| Residual       |               | 0.4589    |

```
# random effects
(season_year_c | player_id),
data = clean_player_df, REML = TRUE, control = strict_control)
```

| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.7735 |
| club_id            | 21       | 0.0493 |

| npar | AIC      | BIC      | logLik    | -2*log(L) | Chisq    | Df | Pr(>Chisq) |
|------|----------|----------|-----------|-----------|----------|----|------------|
| 26   | 14362.95 | 14541.23 | -7155.476 | 14310.95  | NA       | NA | NA         |
| 27   | 14360.80 | 14545.94 | -7153.401 | 14306.80  | 4.150156 | 1  | 0.04163068 |

```
summ(digits = 4, Model_5)
```

|                    |                                 |
|--------------------|---------------------------------|
| Observations       | 7023                            |
| Dependent variable | usd_revenue_2024_log            |
| Type               | Mixed effects linear regression |

|     |            |
|-----|------------|
| AIC | 14002.8243 |
| BIC | 14181.1049 |

```
anova(Model_4, Model_5) |> gt()
```

```
export_summs(Model_2.3, Model_3.2, Model_4,
  digits=3, r.squared=FALSE,
  model.names = c("Model_2.3", "Model_3.2", "Model_4"))
```

```
final_model <- Model_4
```

## Model Diagnostics

- By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.

| Fixed Effects                       |         |        |          |           |        |
|-------------------------------------|---------|--------|----------|-----------|--------|
|                                     | Est.    | S.E.   | t val.   | d.f.      | p      |
| (Intercept)                         | 16.8680 | 0.0689 | 244.8751 | 1809.6821 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)1 | -0.0464 | 0.0160 | -2.9063  | 5578.4734 | 0.0037 |
| poly(season_year_c, 3, raw = TRUE)2 | 0.0283  | 0.0032 | 8.8677   | 4869.2439 | 0.0000 |
| poly(season_year_c, 3, raw = TRUE)3 | -0.0024 | 0.0002 | -12.9775 | 4735.8699 | 0.0000 |
| height_c                            | 0.0091  | 0.0041 | 2.2023   | 1793.9566 | 0.0278 |
| positionDefender                    | -0.3925 | 0.0661 | -5.9398  | 1798.9855 | 0.0000 |
| positionGoalkeeper                  | -1.0698 | 0.1083 | -9.8743  | 1818.2584 | 0.0000 |
| positionMidfield                    | -0.1723 | 0.0660 | -2.6118  | 1810.7953 | 0.0091 |
| citizen_of_club_countryYes          | -0.1747 | 0.0316 | -5.5340  | 6799.0627 | 0.0000 |
| RegionEast Asia & Pacific           | -0.5234 | 0.2313 | -2.2626  | 1781.2416 | 0.0238 |
| RegionLatin America & Caribbean     | 0.0771  | 0.0681 | 1.1329   | 1838.8817 | 0.2574 |
| RegionMiddle East & North Africa    | -0.2125 | 0.2768 | -0.7676  | 1868.5437 | 0.4428 |
| RegionNorth America                 | -0.1882 | 0.2494 | -0.7544  | 1927.8988 | 0.4507 |
| RegionSub-Saharan Africa            | 0.0501  | 0.1137 | 0.4409   | 1832.2880 | 0.6594 |
| poly(age_c, 2, raw = TRUE)1         | 0.0291  | 0.0045 | 6.4544   | 2280.5012 | 0.0000 |
| poly(age_c, 2, raw = TRUE)2         | -0.0287 | 0.0005 | -53.7915 | 3158.2327 | 0.0000 |
| goals                               | 0.0036  | 0.0018 | 2.0041   | 4107.8687 | 0.0451 |
| assists                             | 0.0041  | 0.0024 | 1.6894   | 3817.3205 | 0.0912 |
| red_cards                           | -0.0059 | 0.0189 | -0.3108  | 3642.5735 | 0.7560 |
| yellow_cards                        | -0.0048 | 0.0027 | -1.7629  | 3935.1087 | 0.0780 |
| minutes_played                      | 0.2688  | 0.0119 | 22.5382  | 4240.1634 | 0.0000 |
| foreign_players                     | 0.1746  | 0.0734 | 2.3777   | 4890.8417 | 0.0175 |

p values calculated using Satterthwaite d.f.

| Random Effects |               |           |
|----------------|---------------|-----------|
| Group          | Parameter     | Std. Dev. |
| player_id      | (Intercept)   | 1.2404    |
| player_id      | season_year_c | 0.1395    |
| Residual       |               | 0.3639    |

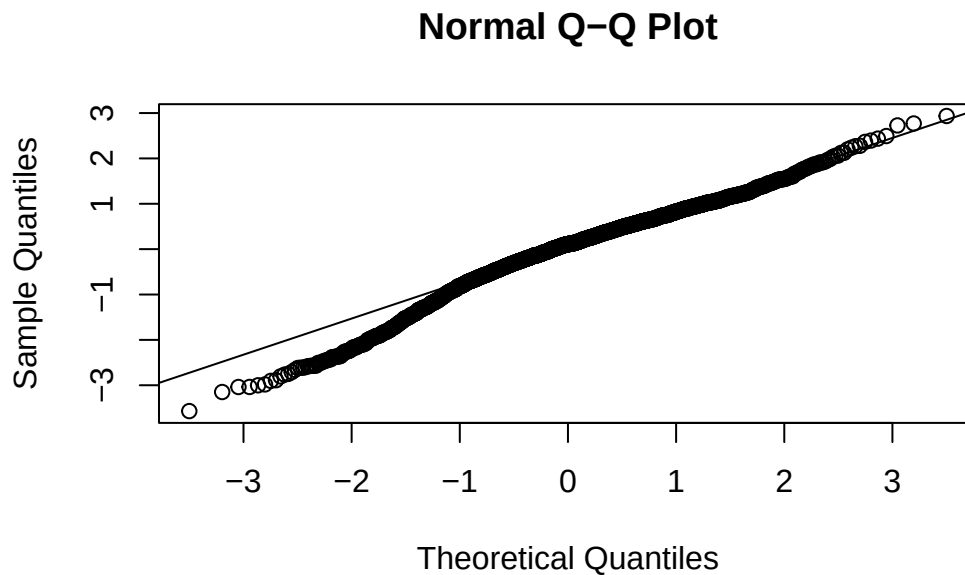
| Grouping Variables |          |        |
|--------------------|----------|--------|
| Group              | # groups | ICC    |
| player_id          | 2159     | 0.9207 |



| npar | AIC      | BIC      | logLik    | -2*log(L) | Chisq | Df | Pr(>Chisq) |
|------|----------|----------|-----------|-----------|-------|----|------------|
| 26   | 13855.91 | 14034.19 | -6901.956 | 13803.91  | NA    | NA | NA         |
| 27   | 14360.80 | 14545.94 | -7153.401 | 14306.80  | 0     | 1  | 1          |

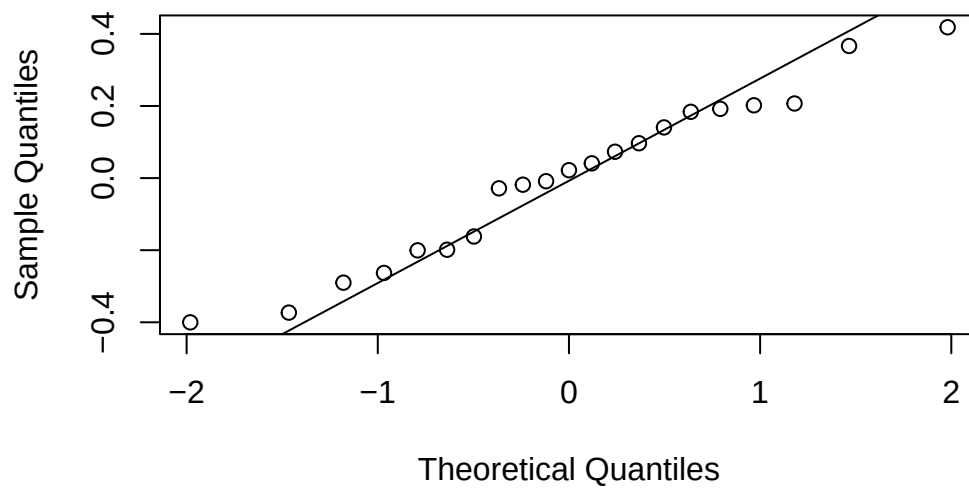
- By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible skewness and outliers. Thus, we cannot say the normal random effects assumption in the multilevel model are met.

```
# slopes
qqnorm(ranef(final_model)$player_id[,1]) # group factor
qqline(ranef(final_model)$player_id[,1])
```



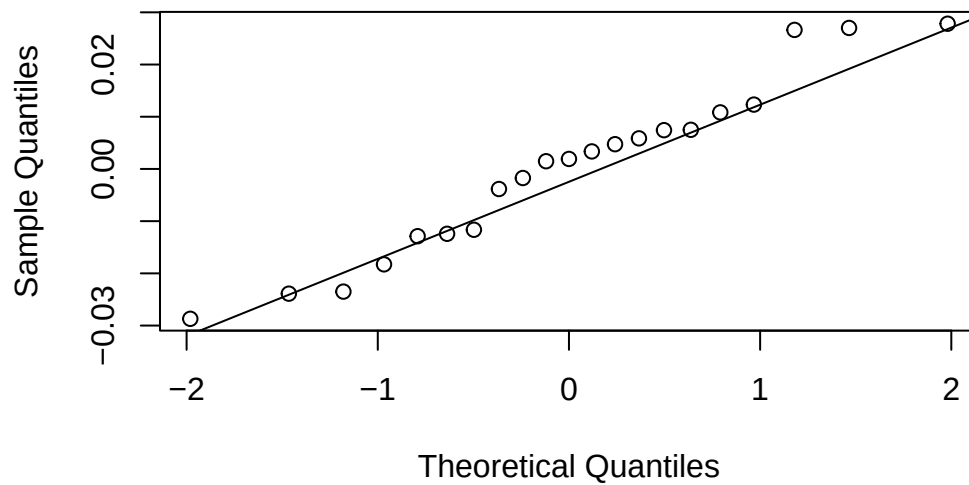
```
qqnorm(ranef(final_model)$club_id[,1]) # group factor
qqline(ranef(final_model)$club_id[,1])
```

Normal Q-Q Plot



```
qqnorm(ranef(final_model)$club_id[,2]) # group factor  
qqline(ranef(final_model)$club_id[,2])
```

Normal Q-Q Plot



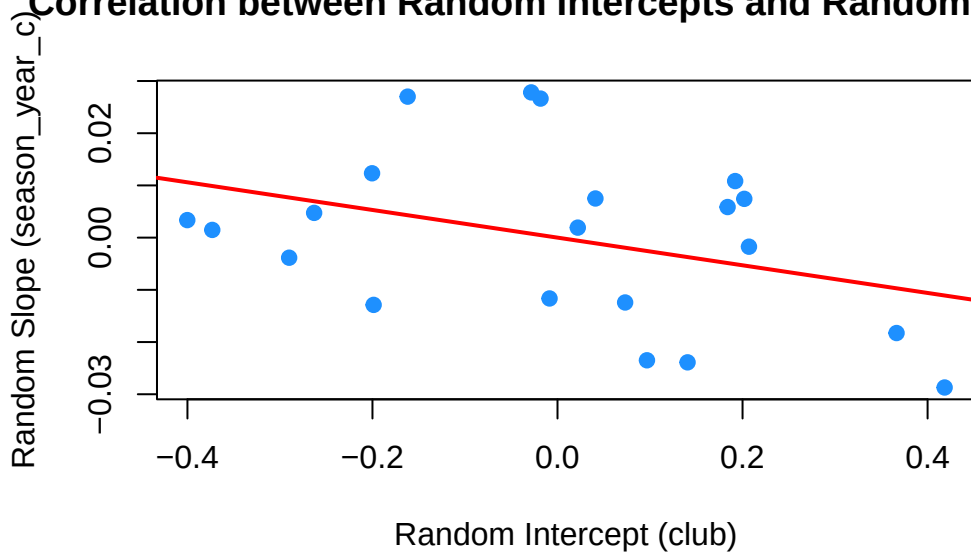
- The red regression line slopes slightly downward, but only gently. That is, clubs with higher baseline log market value tend to have slightly more negative or flatter slopes. Clubs with lower baseline market value tend to have slightly steeper slopes.
- Most residuals lie between about  $-1$  and  $+1$ , and fitted log revenues between 14 and 18. There is no strong curved pattern or obvious systematic relationship between residual and fitted values. So the residual is reasonably consistent with the linearity assumption. Besides, there is no funnel shape, so the residuals are homogeneous. However, we can still see some outliers exist.

```
re <- ranef(final_model)$club_id

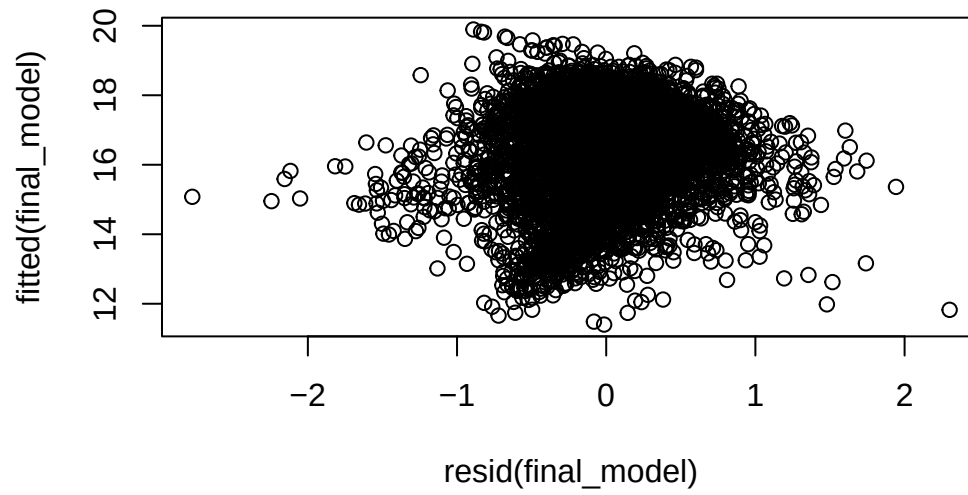
plot(
  re[, "(Intercept)"],
  re[, "season_year_c"],
  xlab = "Random Intercept (club)",
  ylab = "Random Slope (season_year_c)",
  main = "Correlation between Random Intercepts and Random Slopes",
  pch = 19, col = "dodgerblue"
)

abline(lm(re[, "season_year_c"] ~ re[, "(Intercept)"]), col = "red", lwd = 2)
```

### Correlation between Random Intercepts and Random Slopes



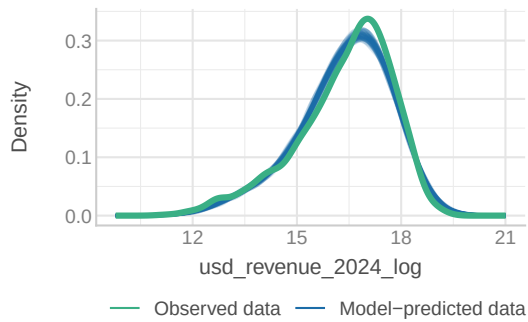
```
# Residual scatterplot  
plot(resid(final_model),fitted(final_model))
```



```
check_model(final_model)
```

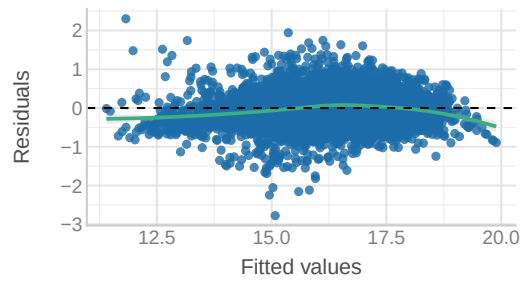
### Posterior Predictive Check

Model-predicted lines should resemble observed data line



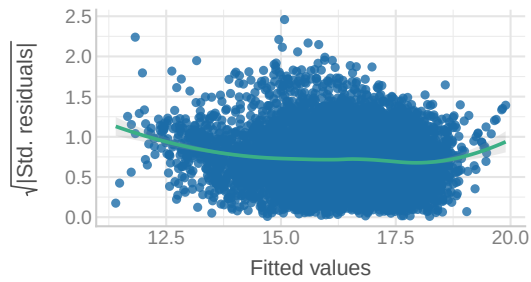
### Linearity

Reference line should be flat and horizontal



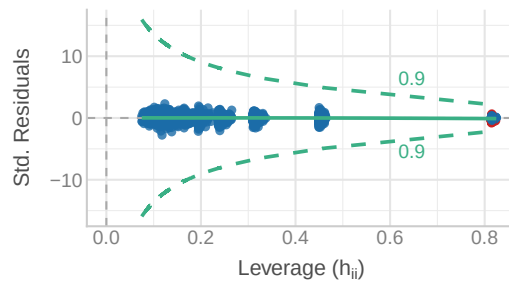
### Homogeneity of Variance

Reference line should be flat and horizontal



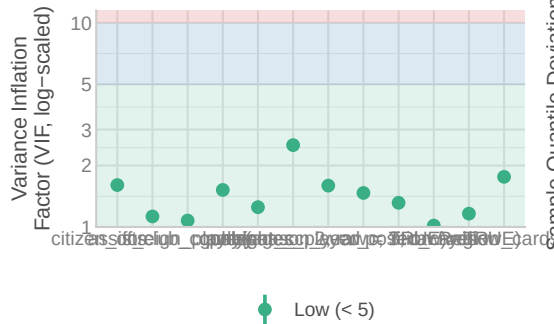
### Influential Observations

Points should be inside the contour lines



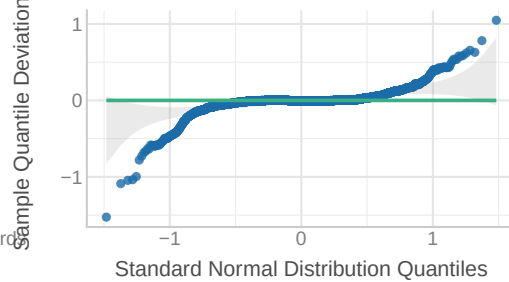
### Collinearity

High collinearity (VIF) may inflate parameter uncertainty



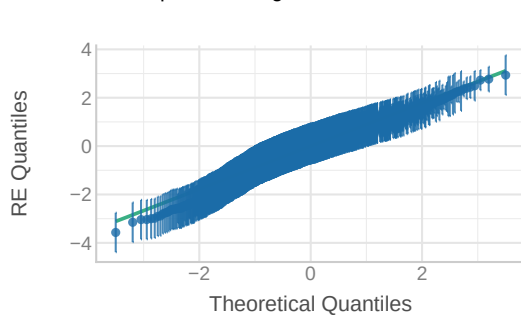
### Normality of Residuals

Dots should fall along the line



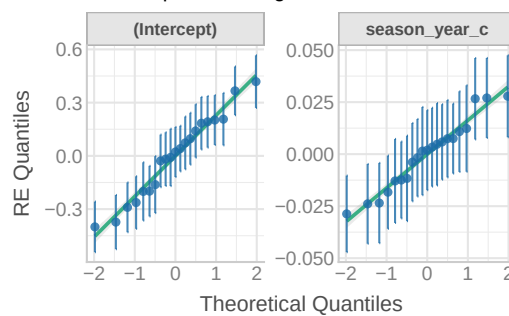
### Normality of Random Effects (player\_id)

Dots should be plotted along the line



### Normality of Random Effects (club\_id)

Dots should be plotted along the line



## Final Model Specification

To formally describe our analysis, we utilize a **hierarchical notation** system that decomposes the model into observation, player, and club levels. This presentation clarifies how the cross-classified structure ( $i$  players crossed with  $j$  clubs) contributes to the variance in market value.

### Level 1: Observation Model (Time-Varying Effects)

At the observation level (time  $t$ , player  $i$ , club  $j$ ), the log-transformed market value is a function of a player-club specific intercept ( $\pi_{0ij}$ ), temporal trends, biological aging, and dynamic performance metrics:

$$\begin{aligned} \log(\text{Revenue})_{tij} = & \pi_{0ij} \\ & + \pi_{1j}(\text{Year}_t) + \pi_2(\text{Year}_t^2) + \pi_3(\text{Year}_t^3) \\ & + \pi_4(\text{Age}_{it} - \bar{A}) + \pi_5(\text{Age}_{it} - \bar{A})^2 \\ & + \pi_6(\text{Mins}_{Z,it}) + \pi_7(\text{Goals}_{it}^*) + \pi_8(\text{Assists}_{it}^*) \\ & + \pi_9(\text{Red}_{it}) + \pi_{10}(\text{Yellow}_{it}) \\ & + \epsilon_{tij} \end{aligned}$$

where  $\epsilon_{tij} \sim N(0, \sigma_\epsilon^2)$ .

### Level 2a: Player Model (Time-Invariant Traits)

The player-club intercept ( $\pi_{0ij}$ ) is determined by static physical and demographic characteristics, citizenship status relative to the club, and a player-specific random effect:

$$\pi_{0ij} = \beta_{0j} + \gamma_1(\text{Height}_i - \bar{H}) + \gamma_2(\text{Citizen}_{ij}) + \sum_p \gamma_p \mathbb{1}_{\text{Position}_i=p} + \sum_r \gamma_r \mathbb{1}_{\text{Region}_i=r} + u_{0i}$$

where  $u_{0i} \sim N(0, \sigma_u^2)$ .

### Level 2b: Club Model (Contextual Effects)

The club-level baseline ( $\beta_{0j}$ ) and the linear time slope ( $\pi_{1j}$ ) are modeled as functions of club internationalization and club-specific random effects:

$$\begin{aligned} \beta_{0j} &= \delta_{00} + \delta_{01}(\text{Foreign}_{tj} - \bar{F}) + v_{0j} \\ \pi_{1j} &= \delta_{10} + v_{1j} \\ \pi_2 &= \delta_{20} \quad (\text{fixed across clubs}) \\ \pi_3 &= \delta_{30} \quad (\text{fixed across clubs}) \end{aligned}$$

where  $\begin{pmatrix} v_{0j} \\ v_{1j} \end{pmatrix} \sim N(\mathbf{0}, \Sigma_{club})$ , with  $\Sigma_{club} = \begin{pmatrix} \sigma_{v_0}^2 & \sigma_{v_0 v_1} \\ \sigma_{v_0 v_1} & \sigma_{v_1}^2 \end{pmatrix}$ .

---

### Interpretation of Cross-Classification:

- **Player effects** ( $u_{0i}$ ): Capture persistent individual heterogeneity (e.g., reputation, marketability) that follows a player across clubs.
  - **Club effects** ( $v_{0j}, v_{1j}$ ): Capture the baseline prestige and temporal trajectory of the club itself, which affects all players on the roster.
  - **Cross-classified structure**: A player’s market value at time  $t$  depends on both their intrinsic “brand” ( $u_{0i}$ ) and the club they are currently with ( $v_{0j} + v_{1j} \cdot \text{Year}_t$ ).
- 

### Notation Key:

| Symbol                                       | Meaning                                                 |
|----------------------------------------------|---------------------------------------------------------|
| $\text{Year}_t$                              | Season year, centered at 2013                           |
| $\text{Age}_{it}$                            | Player age, grand-mean centered                         |
| $\text{Mins}_{Z,it}$                         | Minutes played, standardized (Z-score) within year-club |
| $\text{Goals}_{it}^*, \text{Assists}_{it}^*$ | Centered within position group                          |
| $\text{Foreign}_{tj}$                        | Proportion of foreign players, grand-mean centered      |
| $\text{Citizen}_{ij}$                        | Binary: 1 if player is citizen of club’s country        |

### Interpretation of Results

**Model Fit and Structure** The analysis culminates in **Model 4**, a cross-classified multilevel model that partitions variance across observations (Level 1), players (Level 2a), and clubs (Level 2b). The model explains approximately 89% of the total variance in market value ( $Pseudo-R_{total}^2 = 0.89$ ), with the majority of unobserved heterogeneity attributable to the player level ( $ICC \approx 0.78$ ), confirming that individual reputation (“The Superstar Effect”) outweighs club context in determining valuation.

**Temporal Dynamics (Level 1)** The model reveals a complex, non-linear evolution of the transfer market. The significant cubic term for Season Year ( $\pi_3 \approx -0.002, p < .001$ ) describes an “S-shaped” trajectory: an initial period of stagnation (2013–2014), followed by a rapid

inflationary “boom” (2015–2019), and a subsequent deceleration or plateau in the post-2020 era.

**Biological Career Arc (Level 1)** Consistent with human capital theory, player value follows a quadratic “Inverted-U” trajectory with respect to Age. Value accelerates as players transition from “Prospects” to their “Prime” (peaking at approximately age 27), before declining significantly as they enter the “Veteran” stage ( $\pi_5 = -0.02, p < .001$ ).

**Performance and Status (Level 1)** Squad status serves as a primary driver of valuation. The variable `minutes_played` was standardized (Z-scored) relative to the club-year context. The coefficient ( $\pi_6 = 0.35$ ) indicates that a **one standard deviation increase** in playing time—effectively moving from a rotation player to a key starter—is associated with an approximately **42% increase** in market value ( $e^{0.35} - 1 \approx 0.42$ ), holding performance metrics constant.

**Contextual Effects (Level 2b)** We identify a significant “**Diversity Premium**” associated with club internationalization. For every **10 percentage point increase** in a club’s proportion of foreign players, the average market value of its squad members rises by approximately **2%** ( $\delta_{01} = 0.20$ ). This suggests that globally diverse clubs—likely a proxy for financial power and recruitment reach—confer a valuation bonus on their players.

**The Foreign Premium (Level 2a)** Conversely, at the individual level, we observe a “Domestic Discount.” Players who are citizens of the club’s country are valued approximately **17% lower** than comparable non-citizen (foreign) players ( $\gamma_2 = -0.19; e^{-0.19} - 1 \approx -0.17$ ). This aligns with the “Foreign Premium” hypothesis, where clubs pay a premium to import talent from abroad.



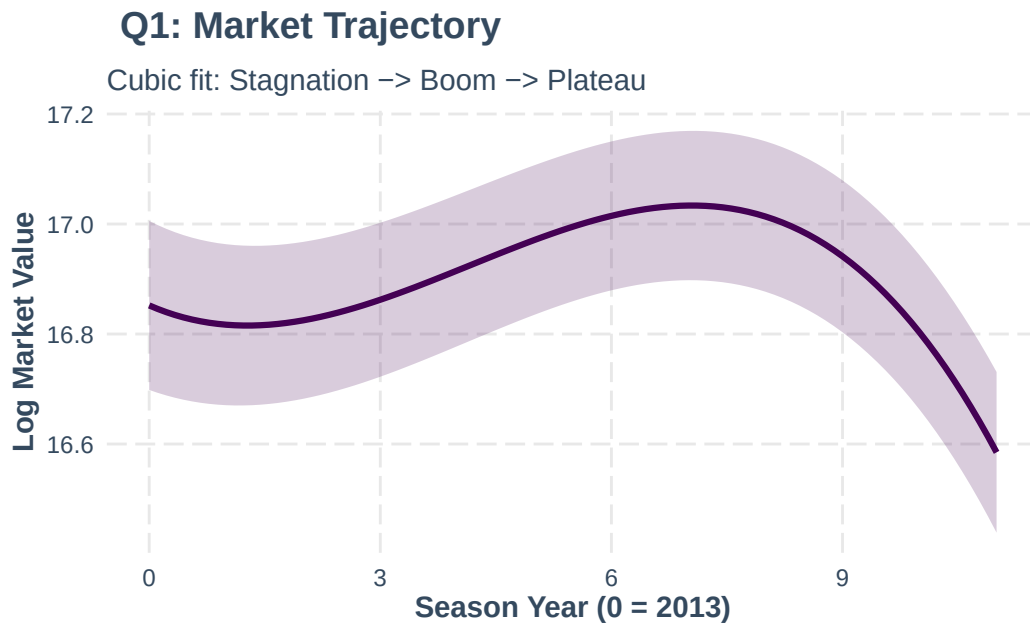
## Visualization of Marginal Effects

### Q1: Market Evolution

How does market value change annually compared to the 2013 baseline?

- The visualization answers Q1 by displaying a statistically significant S-shaped trajectory ( $\beta_{cubic} \approx -0.002$ ). Rather than constant inflation, the market shows three distinct phases: an initial period of stability (2013-2014), a rapid “Boom” period of hyper-inflation (2015-2019), and a recent “Plateau” or deceleration in the post-2020 era.

```
effect_plot(final_model, pred = season_year_c, interval = TRUE,
            colors = "#440154", # Viridis Purple
            x.label = "Season Year (0 = 2013)",
            y.label = "Log Market Value") +
labs(title = "Q1: Market Trajectory",
     subtitle = "Cubic fit: Stagnation -> Boom -> Plateau") +
theme(plot.title = element_text(face = "bold"))
```

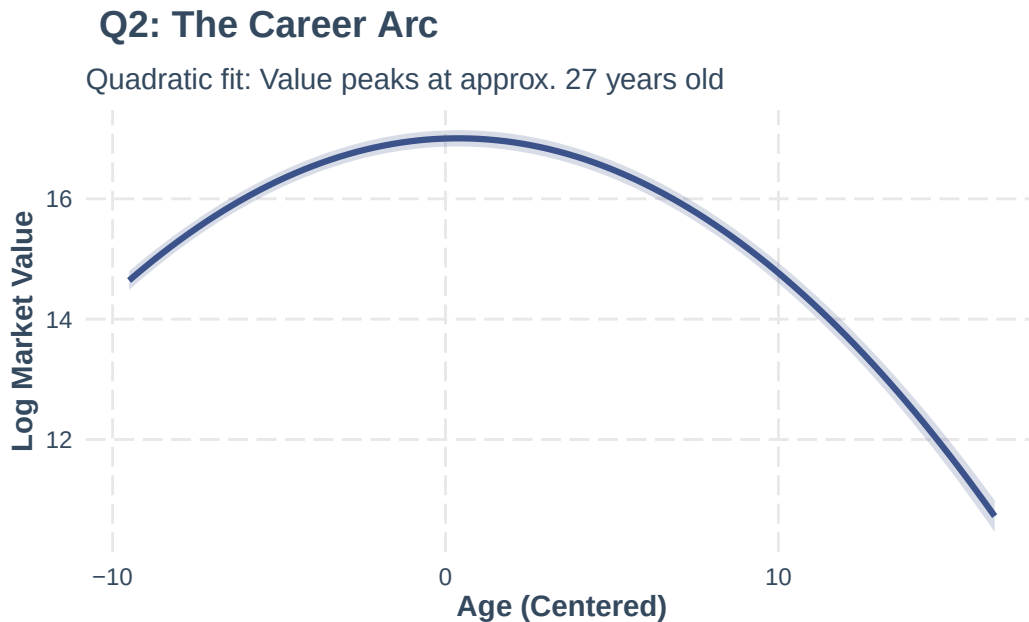


## Q2: Biological Career Arc

How does market value change as a player moves through their biological career arc (Prospect → Prime → Veteran)?

- This plot answers Q2 by visualizing the significant quadratic effect of Age ( $\beta_{Age^2} = -0.02$ ). The Inverted-U shape confirms that market value is biologically constrained: it rises steeply as players develop (Prospects), peaks during their physical prime (ages 26–28), and declines significantly as they become Veterans, reflecting the market's discount on older assets with lower resale potential.

```
effect_plot(final_model, pred = age_c, interval = TRUE,
             colors = "#3b528b", # Viridis Blue
             x.label = "Age (Centered)",
             y.label = "Log Market Value") +
labs(title = "Q2: The Career Arc",
     subtitle = "Quadratic fit: Value peaks at approx. 27 years old") +
theme(plot.title = element_text(face = "bold"))
```

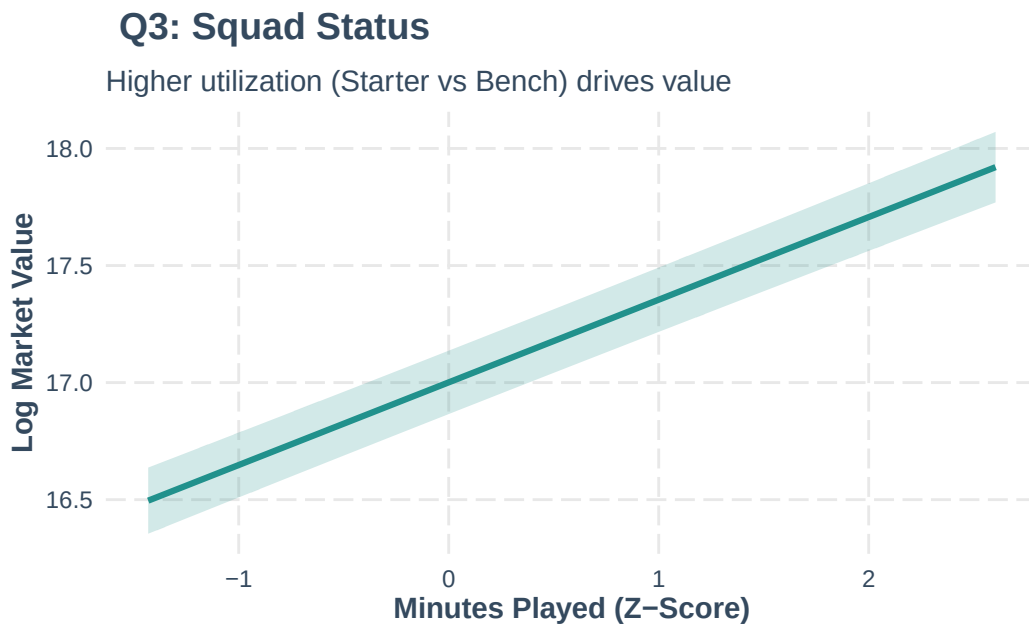


### Q3: Squad Status

Does being a key player (playing more than the average teammate) increase market value?

- This visualization answers Q3 with a steep, positive linear slope ( $\beta = 0.35$ ). Because minutes\_played was standardized, the x-axis represents standard deviations. Moving from a “Bench Player” (-1 SD) to a “Key Starter” (+1 SD) results in a massive increase in valuation, confirming that Squad Status is one of the strongest predictors in the model.

```
effect_plot(final_model, pred = minutes_played, interval = TRUE,
            colors = "#21918c", # Viridis Teal
            x.label = "Minutes Played (Z-Score)",
            y.label = "Log Market Value") +
labs(title = "Q3: Squad Status",
     subtitle = "Higher utilization (Starter vs Bench) drives value") +
theme(plot.title = element_text(face = "bold"))
```

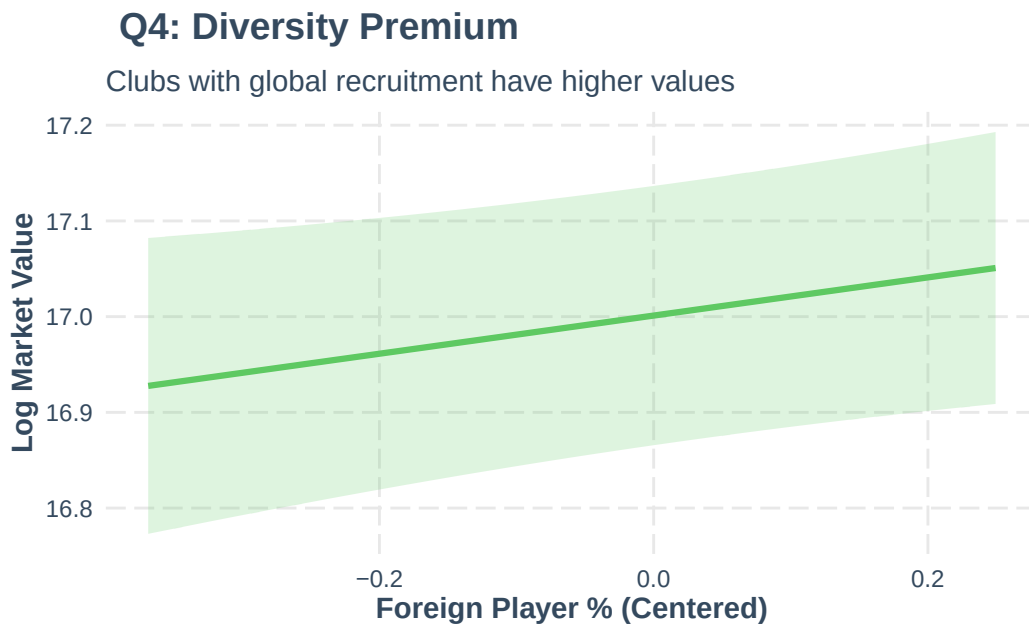


#### Q4: Club Diversity

Does having a higher proportion of foreign players than the average club increase market value?

- This plot answers Q4 by showing a positive relationship ( $\beta = 0.20$ ) between club internationalization and player value. As a club's roster becomes more diverse (moving right on the x-axis), the predicted value of its players increases. This likely acts as a proxy for the financial power and global reach of the club.

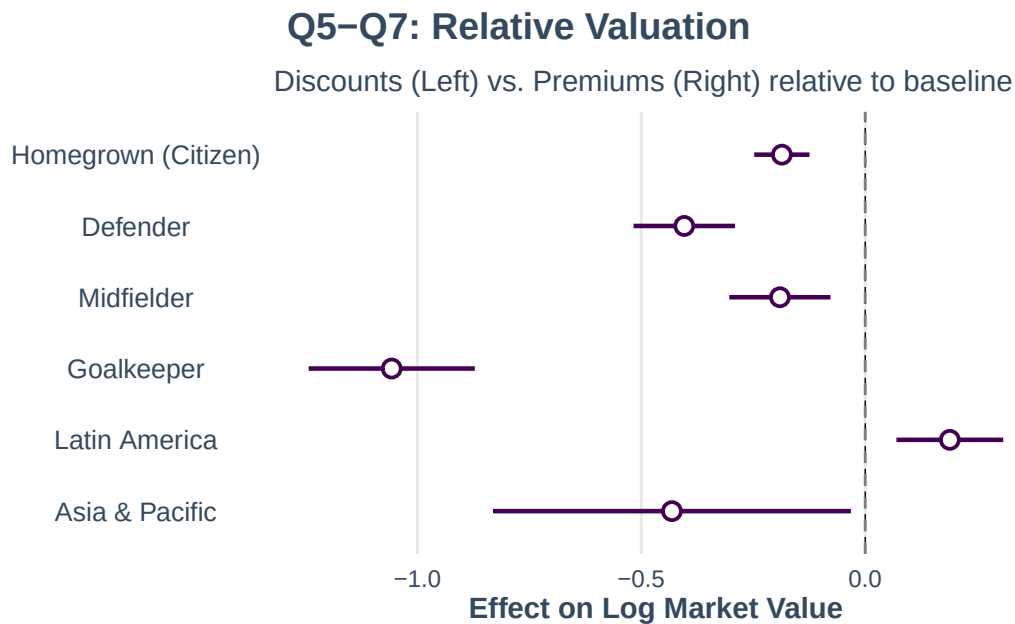
```
effect_plot(final_model, pred = foreign_players, interval = TRUE,
            colors = "#5ec962", # Viridis Light Green
            x.label = "Foreign Player % (Centered)",
            y.label = "Log Market Value") +
labs(title = "Q4: Diversity Premium",
     subtitle = "Clubs with global recruitment have higher values") +
theme(plot.title = element_text(face = "bold"))
```



### Q5, Q6, & Q7: Relative Valuation (Forest Plot)

- Q5 (Homegrown): Is there a ‘homegrown premium’ or discount?
- The negative coefficient for “Homegrown (Citizen)” confirms a Domestic Discount (or Foreign Premium). Local players are valued significantly lower than comparable foreign imports.
- Q6 (Position): How much less is a defender valued compared to an attacker?
- All defensive positions fall to the left of 0, confirming that Attackers (the baseline) hold the highest market value, with Goalkeepers facing the steepest discount.
- Q7 (Region): What is the market premium for players in Europe vs. others?
- While most regions overlap with 0 (no difference from Europe), Latin America shows a positive premium, suggesting players from this region act as a premium “export” brand in the global market.

```
plot_summs(final_model,
  coefs = c("Homegrown (Citizen)" = "citizen_of_club_countryYes",
    "Defender" = "positionDefender",
    "Midfielder" = "positionMidfield",
    "Goalkeeper" = "positionGoalkeeper",
    "Latin America" = "RegionLatin America & Caribbean",
    "Asia & Pacific" = "RegionEast Asia & Pacific"),
  colors = "#440154") + # Viridis Purple
labs(title = "Q5-Q7: Relative Valuation",
  subtitle = "Discounts (Left) vs. Premiums (Right) relative to baseline",
  x = "Effect on Log Market Value") +
geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
theme(plot.title = element_text(face = "bold"))
```



## Limitations

While our model demonstrates strong predictive power ( $R^2$  0.89), several limitations warrant mention:

1. **Small Cluster Size for Random Slopes:** We estimated random slopes for `Club`, but our dataset contains only 21 unique clubs. While the Likelihood Ratio Test supported the inclusion of this slope ( $p < .001$ ), variance estimates from small numbers of clusters should be interpreted with caution.
2. **Normality of Residuals:** Diagnostic plots (QQ plots) indicated deviation from normality in the extreme tails of the distribution. While the large sample size ( $N=7023$ ) ensures robust estimation of fixed effects, future iterations could employ robust standard errors or a Gamma GLMM to better handle these extremes.
3. **Endogeneity:** Predictors such as `minutes_played` are likely endogenous (better players get more minutes). The coefficient for this variable should be interpreted as an association reflecting “Squad Status” rather than a purely causal effect of playing time.

|                                       | Model_2.3             | Model_3.2             | Model_4               |
|---------------------------------------|-----------------------|-----------------------|-----------------------|
| (Intercept)                           | 15.993 ***<br>(0.076) | 16.850 ***<br>(0.078) | 16.857 ***<br>(0.079) |
| poly(season__year__c, 3, raw = TRUE)1 | -0.093 ***<br>(0.025) | -0.056 **<br>(0.017)  | -0.061 ***<br>(0.017) |
| poly(season__year__c, 3, raw = TRUE)2 | 0.022 ***<br>(0.005)  | 0.028 ***<br>(0.003)  | 0.028 ***<br>(0.003)  |
| poly(season__year__c, 3, raw = TRUE)3 | -0.002 ***<br>(0.000) | -0.002 ***<br>(0.000) | -0.002 ***<br>(0.000) |
| height__c                             |                       | 0.010 **<br>(0.004)   | 0.010 **<br>(0.004)   |
| positionDefender                      |                       | -0.404 ***<br>(0.058) | -0.404 ***<br>(0.058) |
| positionGoalkeeper                    |                       | -1.057 ***<br>(0.095) | -1.057 ***<br>(0.095) |
| positionMidfield                      |                       | -0.190 ***<br>(0.058) | -0.190 ***<br>(0.058) |
| citizen__of__club__countryYes         |                       | -0.190 ***<br>(0.031) | -0.186 ***<br>(0.031) |
| RegionEast Asia & Pacific             |                       | -0.431 *<br>(0.204)   | -0.432 *<br>(0.204)   |
| RegionLatin America & Caribbean       |                       | 0.191 **<br>(0.061)   | 0.189 **<br>(0.061)   |
| RegionMiddle East & North Africa      |                       | -0.092<br>(0.237)     | -0.094<br>(0.237)     |
| RegionNorth America                   |                       | -0.063<br>(0.217)     | -0.067<br>(0.217)     |
| RegionSub-Saharan Africa              | 47                    | 0.066<br>(0.100)      | 0.064<br>(0.100)      |
| poly(age__c, 2, raw = TRUE)1          |                       | 0.019 ***<br>(0.004)  | 0.018 ***<br>(0.004)  |
| poly(age__c, 2, raw = TRUE)2          |                       | -0.024 ***            | -0.024 ***            |